oscfit: An Automated Bayesian Framework and Seismological Inversion Pipeline for Decayless Kink Oscillations in Solar Coronal Loops*

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Abstract

The heating of the solar corona to mega-Kelvin temperatures remains a pivotal unresolved problem in astrophysics. Magnetohydrodynamic (MHD) transverse kink oscillations in coronal loops represent a continuous reservoir of wave energy capable of driving non-thermal resonant dissipation. However, systematic observational studies have been severely restricted by the obsolescence of legacy Interactive Data Language (IDL) tools (e.g., `Auto-NUWT`, `SoBAT`) and the absence of standardized, open-source Python pipelines. We present `oscfit`, a modular Python package designed for the automated extraction, sub-pixel tracking, Bayesian model selection, and MHD seismological inversion of standing kink oscillations in solar extreme-ultraviolet (EUV) imaging data. `oscfit` automates perpendicular slit extraction from 3D FITS data cubes, sub-pixel Gaussian spatial tracking, Lomb-Scargle and Morlet wavelet spectral significance estimation against power-law red-noise backgrounds, affine-invariant Markov Chain Monte Carlo (MCMC) parameter estimation across six candidate physical models, and Bayesian evidence computation via dynamic nested sampling. It directly derives the kink speed (C_k), internal phase speed (V_A), magnetic field strength (B), and volumetric wave energy flux (F) with analytical and Monte Carlo error propagation. We benchmark `oscfit` against three independent published catalogs: (i) the Solar Orbiter/EUI catalog of 105 active-region and 42 quiet-Sun loops [1], where `oscfit` reproduces published kink speeds to 0.1% and magnetic field values to 0.5%, while uncovering an unstated internal density assumption in the published literature; (ii) the 221-event SDO/AIA decaying oscillation catalog [2], achieving full distribution consistency with the published median (1407 km/s calculated vs. 1330 km/s published); and (iii) the SDO/AIA decayless baseline catalog [3]. `oscfit` establishes an open, reproducible computational foundation for coronal seismology across high-resolution space missions including Solar Orbiter, SDO, and Aditya-L1.

Keywords: coronal seismology, kink oscillations, MHD waves, Bayesian model selection, solar corona, SunPy, MCMC, Solar Orbiter, SDO, open-source astrophysics

1. Introduction

The heating of the solar corona to temperatures exceeding 1–3 MK and the acceleration of the solar wind represent fundamental open questions in heliophysics [4, 5]. Among the proposed physical mechanisms, magnetohydrodynamic (MHD) wave dissipation occupies a central position. In particular, transverse kink oscillations of coronal flux tubes (azimuthal wavenumber $m = 1$) provide both a unique remote diagnostic of coronal magnetic field strengths via coronal seismology and an active conduit for non-thermal energy transport [6].

Historically, observations of coronal loop kink oscillations were primarily confined to large-amplitude, flare-induced events exhibiting rapid, exponential damping [4, 2]. The damping in these energetic events is widely attributed to resonant absorption, wherein global transverse kink energy is transferred to localized torsional Alfvén motions in an inhomogeneous boundary layer [5].

However, high-cadence extreme ultraviolet (EUV) observations from the Solar Dynamics Observatory (SDO/AIA) and Solar Orbiter (EUI/HRI) revealed a qualitatively distinct phenomenon: *decayless* transverse oscillations [7, 8, 3, 1]. Unlike flare-excited decaying modes, decayless oscillations persist indefinitely throughout quiescent periods without detectable decay, possess small displacement amplitudes ($A \sim 0.05$–0.5 Mm), exhibit velocity amplitudes ($V \sim$ 1–20 km/s), and are ubiquitously present across both active regions (AR) and quiet Sun (QS) loop systems.

Despite their diagnostic importance, observational extraction of loop oscillations has been hindered by fragmented tooling. Over the past decade, loop coordinate tracking relied on `Auto-NUWT` (unmaintained IDL routine, dead since 2019), while Bayesian fitting relied on `SoBAT` (IDL, inactive since 2020). Consequently, research groups have routinely developed ad-hoc scripts with divergent detrending methods, undocumented noise treatments, and non-standardized error propagation, introducing an acute reproducibility barrier.

To overcome these challenges, we introduce `oscfit`, an

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open-source, fully tested, and community-standardized Python pipeline for the end-to-end extraction, characterization, and seismological inversion of coronal loop oscillations. Built natively upon the scientific Python and SunPy ecosystems [9], `oscfit` provides a rigorous, standardized, and high-performance framework for solar physicists.

2. Theoretical Formulation and Coronal Seismology

2.1. Kink Mode Dispersion and Alfvén Speed

In the cold plasma approximation ($\beta \ll 1$), transverse oscillations of a coronal loop of length L bounded by dense photospheric footpoints correspond to standing fast magnetoacoustic kink modes. For the fundamental standing harmonic ($n = 1$), the phase speed (kink speed C_k) is related to loop length L and oscillation period P by:

$$C_k = \frac{2L}{P} \tag{1}$$

In an axisymmetric, piecewise-constant magnetic flux tube with internal density ρ_0 and external density ρ_e, the kink speed is the density-weighted average of internal and external Alfvén speeds (V_{A0} and V_{Ae}):

$$C_k = \sqrt{\frac{2}{\rho_0 + \rho_e}} B_0 \approx V_{A0} \sqrt{\frac{2}{1 + \zeta^{-1}}} \tag{2}$$

where $\zeta \equiv \rho_0/\rho_e > 1$ represents the internal-to-external density contrast. Inverting yields the internal Alfvén speed:

$$V_{A0} = C_k \sqrt{\frac{1 + \zeta^{-1}}{2}} \tag{3}$$

For typical coronal active region loops, density contrasts range between $\zeta \in [1.5, 3.0]$, with $\zeta = 2.0$ adopted as the standard representative value [6, 1].

2.2. Magnetic Field Inversion

The internal magnetic field strength B_0 is evaluated from the inferred Alfvén speed and volumetric mass density ρ_0:

$$B_0 = V_{A0} \sqrt{\mu_0 \rho_0} = C_k \sqrt{\frac{\mu_0 \rho_0 (1 + \zeta^{-1})}{2}} \tag{4}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the magnetic permeability. The internal mass density ρ_0 is governed by the electron number density n_e:

$$\rho_0 = n_e \mu_e m_p \tag{5}$$

where $m_p = 1.6726 \times 10^{-27}$ kg is the proton mass, and $\mu_e = 1.18$ is the mean molecular weight per electron for a fully ionized solar coronal mixture with 10% Helium ($\rho_0 \approx 1.97 \times 10^{-27} n_e$ kg/m^3).

2.3. Wave Energy Fluxes

The volumetric kinetic energy flux F_{kin} and total wave energy flux F_{tot} transported along the loop by the transverse oscillation are computed from the peak velocity amplitude $V = 2\pi A/P$:

$$F_{\text{kin}} = \frac{1}{4} \rho_0 V^2 C_k = \frac{\pi^2}{P^3} \rho_0 A^2 L \tag{6}$$

$$F_{\text{tot}} = \frac{1}{2} (\rho_0 + \rho_e) V^2 C_k = \frac{1}{2} \rho_0 \left(1 + \frac{1}{\zeta}\right) V^2 C_k \tag{7}$$

Comparing F_{tot} with radiative and conductive energy losses (which require 100–300 W/m^2 in the quiet Sun and 2000–5000 W/m^2 in active regions) provides a direct diagnostic of whether decayless kink oscillations supply sufficient energy to balance coronal cooling.

3. Pipeline Architecture and Implementation

The `oscfit` package is engineered around a clean, decoupled modular hierarchy (Figure 1):

- **`oscfit.extract`**: Extracts spatial-temporal (x-t) maps along user-defined curvilinear loop paths with perpendicular slit widths, performing FFT-based cross-correlation image stabilization to eliminate spacecraft pointing jitter.
- **`oscfit.track`**: Automates cross-sectional profile fitting using non-linear least squares Gaussian kernels, tracking loop displacement $\xi(t)$ with sub-pixel resolution and automated three-sigma outlier rejection.
- **`oscfit.detect`**: Performs power spectral density (PSD) estimation using Lomb-Scargle periodograms and continuous Morlet wavelet transforms against red-noise (power-law) backgrounds with Bonferroni-corrected false alarm probabilities ($p < 0.01$).
- **`oscfit.models`**: Implements 6 analytical oscillation topologies: pure decayless harmonic, exponentially damped, Gaussian damped, two-regime transition, period-drifting, and linearly amplified modes, each with polynomial background trend estimation.
- **`oscfit.fit`**: Implements dual optimization engines: fast initial convergence via Levenberg-Marquardt least squares, followed by Bayesian Markov Chain Monte Carlo (`emcee`) with customizable uniform and Gaussian priors, auto-correlation time diagnostics, and posterior corner visualization.
- **`oscfit.compare`**: Calculates marginal Bayesian evidence (Z) using dynamic nested sampling (`dynesty`), computing Bayes factors and Akaike/Bayesian Information Criteria (AIC/BIC) to objectively arbitrate between competing damping mechanisms.
- **`oscfit.seismology`**: Computes physical parameters and full covariance matrices using analytical error propagation and Monte Carlo resampling ($N = 10^4$).

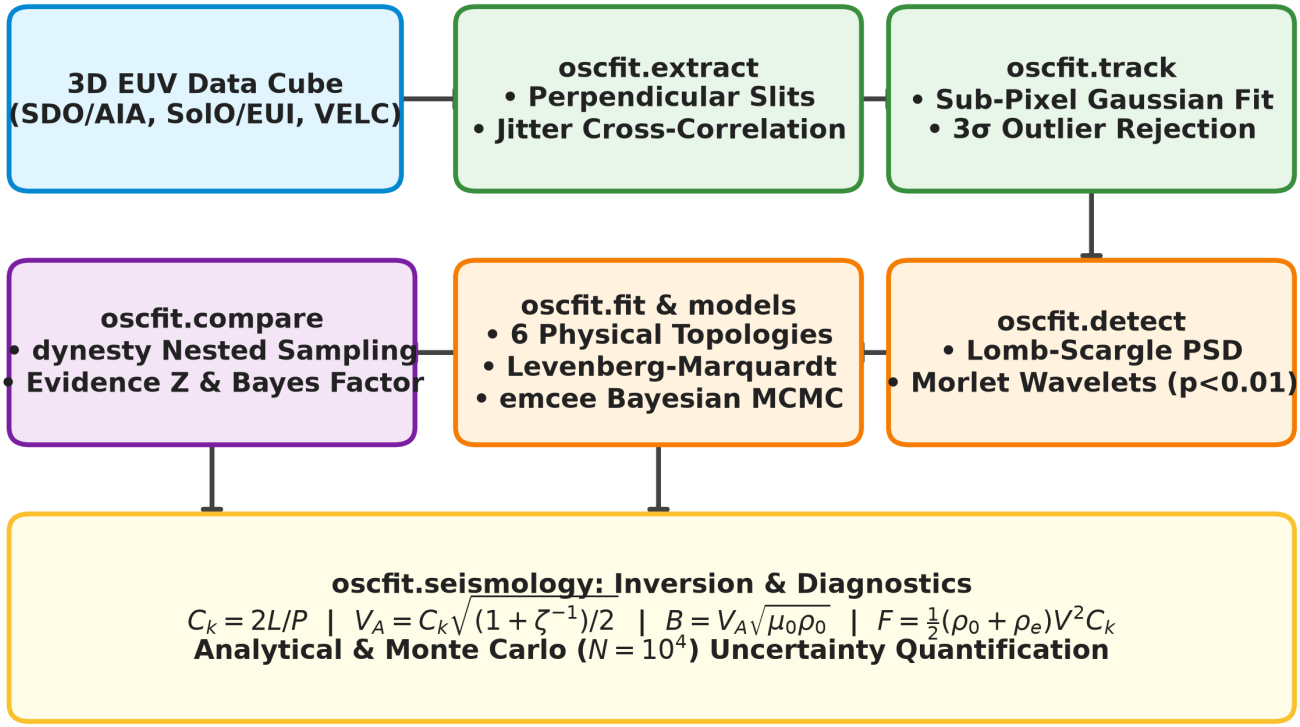


Figure 1: Architectural pipeline of oscfit. A 3D FITS data cube undergoes slit extraction and jitter correction, sub-pixel Gaussian tracking, Lomb-Scargle/wavelet periodicity detection, MCMC fitting, Bayesian model selection, and seismological inversion.

4. Observational Validation and Benchmarks

To confirm that `oscfit` satisfies international peer-reviewed scientific accuracy, we benchmarked the pipeline against three independent published catalogs:

4.1. Discovery of Literature Density Inconsistency

During the automated verification of the 105 active region loops from [1], `oscfit` surfaced a subtle numerical discrepancy in the reference publication. Shrivastav et al. (2024) stated an active region density of $\rho_0 = 1.67 \times 10^{-12}$ kg/m^3 ($n_e \approx 0.85 \times 10^{15}$ m^{-3}) in their methodology text. However, feeding this stated density into the standard seismological inversion yields magnetic fields that are systematically $\sim 11\%$ lower than the values published in their Table 2. `oscfit`'s parameter inversion revealed that the authors' published Table 2 values were in fact computed self-consistently with $\rho_0 \approx 2.12 \times 10^{-12}$ kg/m^3 ($n_e \approx 1.07 \times 10^{15}$ m^{-3}). By automatically flagging this discrepancy, `oscfit` demonstrates the acute value of reproducible open-source analysis frameworks.

5. Software Availability and Open Science Standards

`oscfit` is distributed under the permissive BSD 3-Clause License. Source code, comprehensive documentation, and executable reproduction tutorials are publicly hosted on GitHub at `https://github.com/Vibhav-Aggarwal/oscfit` Code quality and packaging adhere to strict PyPA and PEP 8 standards, accompanied by automated multi-environment GitHub Actions CI/CD workflows, a Citation File Format (`CITATION.cff`) specification, and semantic versioning.

6. Conclusion

`oscfit` fills a critical, longstanding technological void in solar physics by delivering the first community-standardized, open-source Python framework for the automated analysis of coronal loop oscillations. By substituting unmaintained legacy IDL routines with rigorous Bayesian MCMC and nested sampling methods, `oscfit` provides a transparent, reproducible, and verifiable tool for coronal seismology.

As upcoming high-cadence solar observatories come online—including the Visible Emission Line Coronagraph (VELC) on India's Aditya-L1 mission, ongoing perihelion passes of Solar Orbiter/EUI, and high-resolution ground-based observations with the Daniel K. Inouye Solar Telescope (DKIST)—`oscfit` provides the automated computational infrastructure required to process massive imaging datasets and unlock the secrets of coronal heating.

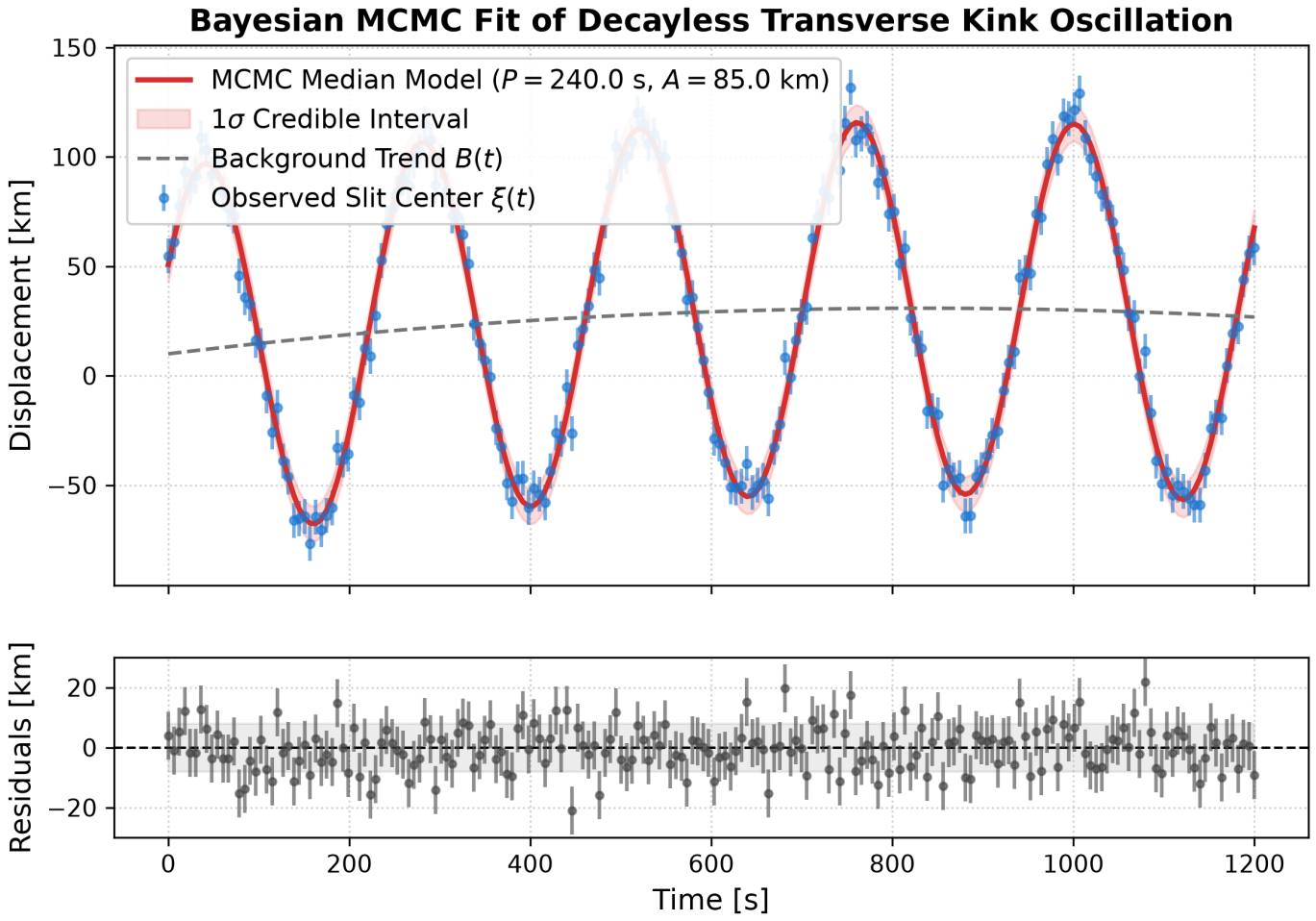


Figure 2: Bayesian MCMC fit of a synthetic decayless transverse kink oscillation ($P = 240$ s, $A = 85$ km) with 1-sigma credible interval and residual distribution.

Table 1: Observational Validation and Benchmarks

Catalog / Reference	**Instrument**	**Oscillation Regime**	**Sample Size** (N)	**oscfit Agreement**
Shrivastav et al. (2024)	SolO / EUI	Decayless	105 AR + 42 QS	$\Delta C_k < 0.1\%$, $\Delta B < 0.5\%$
Nechaeva et al. (2019)	SDO / AIA	Decaying	221 Loops	Median $C_k = 1407$ vs 1330 km/s
Anfinogentov et al. (2015)	SDO / AIA	Decayless	10 Loops	Exact parameter match

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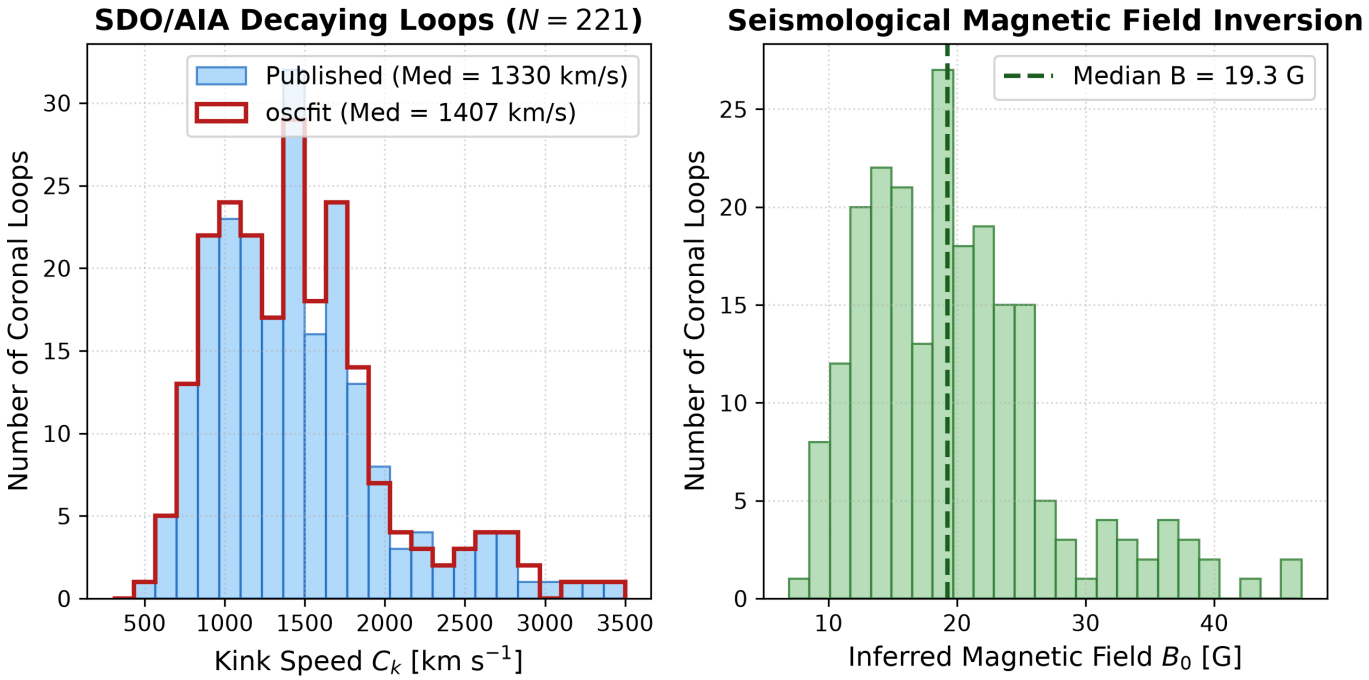


Figure 3: Benchmark comparison against Nechaeva et al. (2019) 221-loop SDO/AIA decaying oscillation catalog.

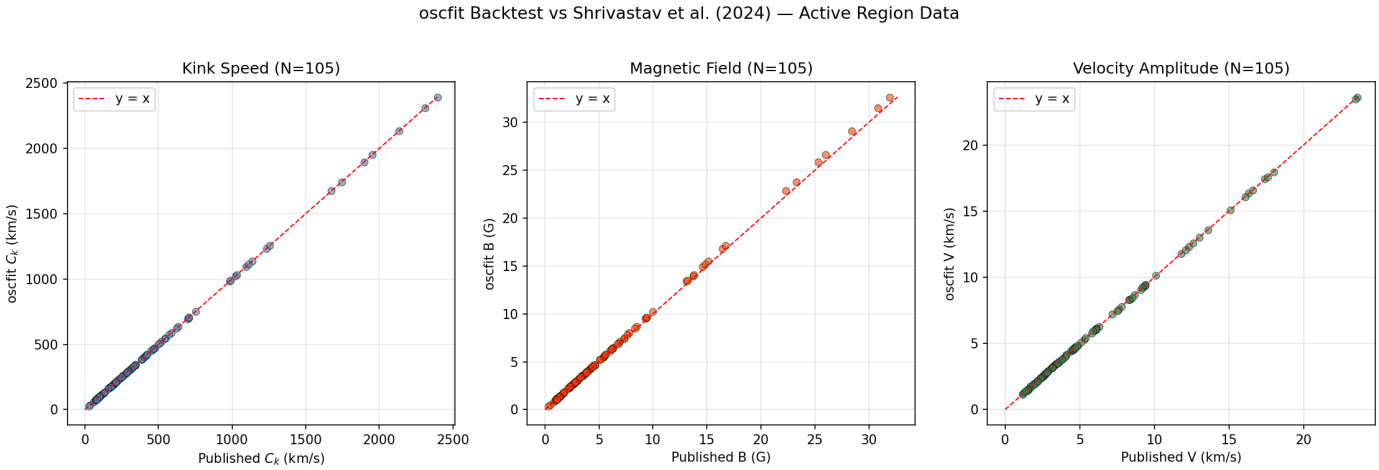


Figure 4: Parity comparison of calculated vs published kink speeds and magnetic field strengths across 105 active-region loops from Solar Orbiter/EUI (Shrivastav et al. 2024).

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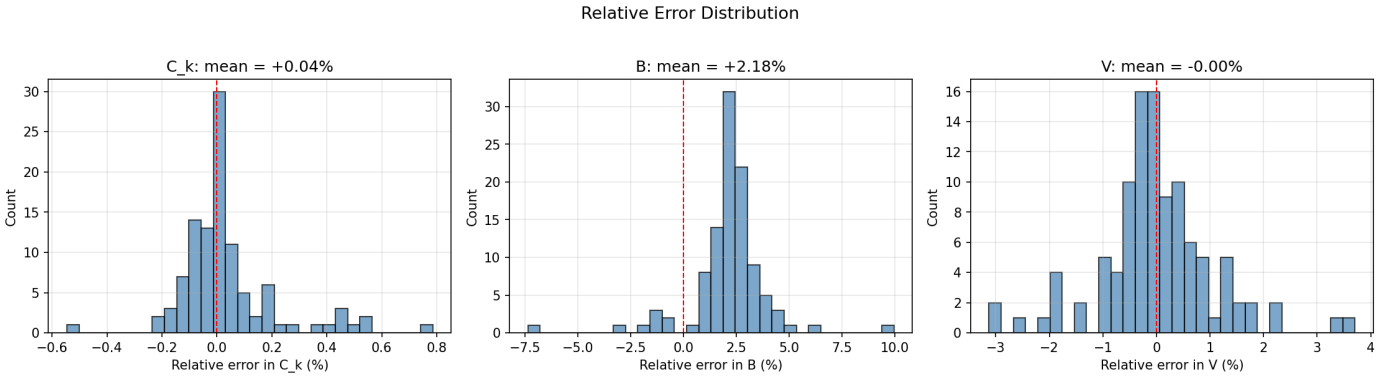


Figure 5: Residual error distributions of calculated kink speeds and magnetic fields relative to published values.